

University of Colombo School of Computing
SCS 3312 – Research Methods

**Study of Passenger Waiting Behaviour
at Thummulla Bus Stop**

D0: Topic Proposal & Research Plan

Group Submission

Group Number: 2

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May 2026

Acknowledgement

We would like to express our sincere gratitude to the faculty of the University of Colombo School of Computing for providing this opportunity to engage with real-world research environments. We also thank the passengers and commuters at the selected bus stops whose cooperation made field observations possible, as well as the University of Colombo administration for facilitating access to the study sites.

Abstract

This proposal presents a research plan to investigate passenger waiting behaviour and boarding decisions at three selected urban bus stops in Colombo, Sri Lanka. The study is motivated by the observable yet under-studied phenomena of passengers extending their waiting duration, rejecting overcrowded buses, and exhibiting irregular boarding patterns within public transport environments. Using a mixed-method approach that combines qualitative techniques—direct observation and informal interviews—with quantitative methods—passenger counting, waiting-time measurement, and structured surveys—the research aims to identify behavioural patterns and the factors that influence them. Three field sites, namely Reid Avenue, Town Hall, and Fort Main bus stops, have been selected because they represent varied levels of passenger congestion and operational conditions. Data will be collected during both peak and off-peak hours to capture the full range of commuter behaviour. The findings are expected to contribute practical insights for improving passenger management and public transport planning in urban Sri Lanka.

Contents

Acknowledgement	i
Abstract	ii
List of Figures	iv
List of Tables	iv
List of Acronyms	1
1 Introduction	2

1.1	Background of the Study	2
1.2	Research Problem	3
1.3	Justification of the Study	3
1.4	Research Topic	4
1.5	Relevance of the Study	4
1.6	Research Objectives	5
1.7	Preliminary Research Questions	5
1.8	Scope of the Study	5
2	Field Study Description	7
2.1	Selected Field Sites	7
2.2	Description of the Selected Bus Stops	7
2.2.1	Reid Avenue Bus Stop	8
2.2.2	Town Hall Bus Stop	8
2.2.3	Fort Main Bus Stop	8
2.3	Peak and Off-Peak Conditions	8
2.3.1	Peak Hours	8
2.3.2	Off-Peak Hours	9
2.4	Observed Passenger Activities	9
3	Research Methodology	10
3.1	Research Approach	10
3.2	Qualitative Data Collection Methods	10
3.2.1	Observation	10
3.2.2	Informal Interviews	11
3.2.3	Field Notes	11
3.3	Quantitative Data Collection Methods	11
3.3.1	Passenger Counting	11
3.3.2	Waiting Time Measurement	12
3.3.3	Survey Data Collection	12
3.4	Variables to be Measured	12
3.5	Data Recording Procedure	12
3.6	Group Member Responsibilities	13
4	Critical Analysis	14
4.1	Importance of Studying Passenger Behaviour	14
4.2	Expected Challenges During Data Collection	14
4.3	Possible Limitations of the Study	15

4.4 Ethical Considerations	15
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List of Figures

List of Tables

List of Acronyms

UCSC	University of Colombo School of Computing
LMS	Learning Management System
SCS	School of Computing Studies

Chapter 1 Introduction

This chapter provides an overview of the research study and introduces the main problem being investigated. It explains the background and importance of studying passenger waiting behaviour at urban bus stops, presents the research objectives and questions, and defines the scope of the study. Furthermore, this chapter establishes the conceptual foundation for understanding how passenger behaviour is shaped by factors such as waiting time, crowd conditions, and bus availability in real-world transport environments.

1.1 Background of the Study

Public transportation is one of the most essential services used by people in urban areas. In Sri Lanka, buses serve as the primary mode of daily travel for students, office workers, and the general public. Consequently, bus stops have become crowded public spaces where passengers wait, interact, and make consequential decisions relating to transportation.

Passenger behaviour at bus stops varies considerably depending on conditions such as waiting time, crowd level, bus availability, and urgency of travel. Some passengers wait for a specific bus route for extended periods, while others board the first available bus. In crowded situations, passengers may avoid overcrowded buses, move frequently within the waiting area, or attempt aggressive boarding when buses arrive.

These behaviours have direct implications for passenger safety, comfort, queue discipline, and the overall efficiency of public transportation services. Studying passenger waiting behaviour therefore offers a window into how people make real-world transport decisions under varying environmental pressures.

This study focuses on observing and analysing passenger waiting behaviour and boarding decisions at three selected urban bus stops in Colombo during both peak and off-peak hours.

1.2 Research Problem

Passenger behaviour at bus stops is frequently influenced by overcrowding, irregular bus arrivals, prolonged waiting times, and individual travel preferences. Although buses may arrive with reasonable frequency, many passengers continue to wait for specific services rather than boarding immediately. Some passengers reject visibly overcrowded buses, while others display visible impatience during extended waits.

At congested bus stops, passengers may also demonstrate behaviours such as queue changing, standing at the roadside, relocating between waiting areas, or rushing to board arriving buses. These behaviours can create safety hazards, increase congestion, and reduce boarding efficiency.

Despite the prevalence of these phenomena, there is limited scholarly understanding of the factors that govern passenger waiting decisions and behavioural responses in urban bus stop environments. This research therefore aims to investigate why passengers extend their waiting periods and how different environmental and social factors affect their behaviour at bus stops.

1.3 Justification of the Study

This problem is worth investigating for several interconnected reasons. First, there is a notable gap in the existing literature regarding passenger waiting behaviour in the Sri Lankan urban transport context. While studies on public transport have been conducted in cities across Europe, East Asia, and North America, comparable empirical research within Colombo's bus network remains sparse. Without locally grounded evidence, transport authorities lack the data needed to address congestion, boarding inefficiencies, and passenger dissatisfaction at bus stops.

Second, the problem carries direct practical significance. Colombo's bus stops serve hundreds of thousands of commuters daily, and observable issues—such as overcrowding, disorganised boarding, and prolonged waiting—affect not only individual passengers but also the operational performance of the entire public transport system. Understanding why passengers make particular waiting and boarding decisions can inform targeted interventions, including improved queue management, better scheduling, and more effective bus stop design.

Third, passenger behaviour at bus stops raises safety concerns that warrant sys-

tematic investigation. Passengers who stand near the roadside, rush to board arriving buses, or engage in aggressive boarding during peak hours are exposed to heightened risk of injury. Identifying the environmental and situational triggers for these behaviours is a necessary step toward developing safer transit environments.

Finally, from a methodological standpoint, the problem offers a rich opportunity to apply and integrate qualitative and quantitative research methods to a real-world phenomenon. Studying observable human behaviour in public spaces provides a practical context for developing competence in mixed-method research design, which is a central objective of this module.

1.4 Research Topic

"Passenger Waiting Behaviour and Boarding Decisions at Selected Urban Bus Stops"

This research examines how passengers behave while waiting for buses, and how factors such as crowd density, waiting duration, and bus conditions influence their boarding decisions.

1.5 Relevance of the Study

Public buses constitute one of the primary transportation modes used in Colombo. Passenger behaviour at bus stops directly affects crowd management, transport efficiency, and passenger safety. Understanding waiting behaviour can help to:

- Identify common behavioural patterns at bus stops;
- Understand the factors that cause prolonged waiting times;
- Improve passenger management strategies;
- Support better public transport planning; and
- Reduce overcrowding and unsafe boarding behaviour.

In addition, this research is relevant to the field of research methods because it studies observable, real-world human behaviour using both qualitative and quantitative approaches, making it a suitable vehicle for developing methodological competence.

1.6 Research Objectives

The primary objective of this study is to investigate passenger waiting behaviour and boarding decisions at selected bus stops in Colombo. The specific objectives are:

1. To identify common passenger waiting behaviours at bus stops;
2. To examine the factors influencing passenger waiting time;
3. To analyse how crowd density affects boarding decisions;
4. To compare passenger behaviour between different bus stops; and
5. To observe behavioural differences during peak and off-peak hours.

1.7 Preliminary Research Questions

This study attempts to address the following research questions:

1. Why do some passengers wait longer than others at bus stops?
2. How does crowd density influence passenger boarding decisions?
3. Why do some passengers reject overcrowded buses?
4. How do passengers behave during extended waiting periods?
5. Are there observable differences in passenger behaviour between different bus stops?
6. How do peak-hour conditions affect passenger waiting behaviour?

1.8 Scope of the Study

This research is limited to observing passenger waiting behaviour and boarding decisions at three selected urban bus stops in Colombo. The study focuses on:

- Passenger waiting time;
- Boarding and rejection behaviour;
- Crowd conditions within waiting areas;
- Passenger movement within bus stops; and
- Peak-hour and off-peak-hour behaviour.

Data will be collected through direct observation, informal interviews, surveys, passenger counting, and waiting-time measurements. The study does not encompass technical bus operations, government transport policies, or driver behaviour.

The primary focus remains on passenger behaviour and decision-making within bus stop environments.

Chapter 2 Field Study Description

This chapter describes the locations selected for the research study and explains the environmental conditions observed at each bus stop. It discusses the differences between peak-hour and off-peak-hour conditions, and identifies common passenger activities and behavioural patterns noted during preliminary field visits. The purpose of this chapter is to provide a clear understanding of the real-world setting in which the research will be conducted.

2.1 Selected Field Sites

The research will be conducted at three urban bus stops in Colombo. These sites were selected because they experience different passenger volumes, bus frequencies, and crowd conditions throughout the day, thereby providing suitable environments for observing a range of passenger behaviours related to waiting, boarding, and decision-making.

The selected field sites are:

- Reid Avenue Bus Stop
- Town Hall Bus Stop
- Fort Main Bus Stop

These locations are commonly used by students, office workers, and the general public. The bus stops also represent different levels of passenger congestion, allowing meaningful comparisons between highly crowded and moderately crowded environments.

2.2 Description of the Selected Bus Stops

Each selected bus stop has distinct physical and environmental characteristics that may influence passenger behaviour.

2.2.1 Reid Avenue Bus Stop

This bus stop is primarily used by university students and nearby office workers. Passenger traffic increases significantly during morning and evening hours. The waiting space is limited, which sometimes causes crowding near the roadside and contributes to disorganised boarding behaviour.

2.2.2 Town Hall Bus Stop

Town Hall Bus Stop is situated in a busy commercial and administrative area. A large number of passengers use this location throughout the day. Due to the high number of bus routes served, passengers frequently move within the waiting area and often change their boarding decisions in response to crowd conditions.

2.2.3 Fort Main Bus Stop

Fort Main Bus Stop experiences very high passenger flow as it functions as a major transport hub in Colombo. Passengers at this location regularly encounter long waiting times, overcrowding, and competition for boarding space during peak hours.

These differences between bus stops provide opportunities to compare passenger behaviour under contrasting environmental and operational conditions.

2.3 Peak and Off-Peak Conditions

Passenger behaviour at bus stops changes substantially depending on the time of day and prevailing crowd conditions. The study will therefore observe both peak-hour and off-peak-hour situations.

2.3.1 Peak Hours

Peak hours typically occur during morning travel periods and evening return hours. During these periods:

- Passenger density increases markedly;
- Waiting areas become congested;
- Boarding competition intensifies; and
- Passengers may become impatient as a result of delays or overcrowding.

2.3.2 Off-Peak Hours

Off-peak periods generally exhibit lower passenger volumes, shorter waiting times, reduced crowd pressure, and more organised boarding behaviour. Observing both conditions will help identify how environmental factors influence waiting behaviour and decision-making.

2.4 Observed Passenger Activities

During preliminary observations, several common passenger activities and behavioural patterns were identified at the selected bus stops. These include:

- Waiting for specific bus routes;
- Rejecting overcrowded buses upon arrival;
- Queue changing and position shifting;
- Moving closer to the roadside when buses approach;
- Using mobile phones while waiting;
- Interacting with fellow passengers;
- Attempting aggressive boarding during congested situations; and
- Frequently checking for approaching buses.

Differences were also observed between passengers who were prepared to wait for extended periods and those who preferred immediate boarding regardless of conditions. These behaviours indicate that passenger decisions are shaped by waiting time, crowd conditions, comfort, and urgency of travel, and they establish a firm foundation for the qualitative and quantitative data collection planned for subsequent phases of the study.

Chapter 3 Research Methodology

This chapter explains the research methods and procedures that will be used to collect and analyse data. It describes both qualitative and quantitative approaches to investigating passenger waiting behaviour and boarding decisions at selected bus stops, and outlines methods of observation, interviews, surveys, data recording, and the responsibilities of each group member.

3.1 Research Approach

This research adopts a mixed-method approach, combining qualitative and quantitative data collection. The mixed-method design was selected because passenger behaviour at bus stops involves both observable behavioural patterns and measurable numerical data (?).

The qualitative strand will illuminate passenger experiences, behaviours, and decision-making processes through direct observation and informal interviews. The quantitative strand will produce measurable data on waiting times, passenger counts, and boarding frequencies. By integrating both strands, the study aims to achieve a more comprehensive understanding of passenger waiting behaviour under varying transport conditions.

3.2 Qualitative Data Collection Methods

The qualitative phase focuses on understanding passenger behaviour, attitudes, and decision-making patterns. Data will be collected through observation, informal interviews, and field notes.

3.2.1 Observation

Direct observation will be used to study passenger activities and behavioural patterns at the selected bus stops during both peak and off-peak hours. Observations will focus on:

- Passenger movement within the bus stop;
- Queue behaviour and queue discipline;
- Boarding behaviour;
- Bus rejection behaviour;
- Reactions to delays and overcrowding; and
- Passenger interactions with others.

Observation allows researchers to study natural passenger behaviour in real-world settings without interfering with normal activities (?).

3.2.2 Informal Interviews

Short informal interviews will be conducted with selected passengers. These interviews will help to clarify the reasons behind observed decisions and behaviours. Topics will include:

- Reasons for waiting for a specific bus;
- Opinions regarding overcrowded buses;
- Factors affecting boarding decisions; and
- Passenger experiences during long waiting periods.

Interviews will be brief, respectful, and conducted without disturbing passengers.

3.2.3 Field Notes

Field notes will record important observations and environmental conditions during data collection, including crowd conditions, weather, unusual passenger behaviours, traffic conditions, and any special incidents observed. Field notes will support the interpretation of qualitative findings in later analytical stages.

3.3 Quantitative Data Collection Methods

The quantitative phase focuses on collecting measurable numerical data related to passenger waiting behaviour.

3.3.1 Passenger Counting

Passenger counting will measure the number of passengers present at each bus stop across different time periods, including the number arriving, boarding, and

rejecting buses, as well as passenger density during peak and off-peak hours. This data will enable identification of crowd patterns and passenger flow.

3.3.2 Waiting Time Measurement

Waiting-time measurements will determine how long passengers remain at bus stops before boarding. Records will capture average and maximum waiting times, differences between peak and off-peak periods, and waiting durations for different bus routes.

3.3.3 Survey Data Collection

Structured questionnaires will be distributed to selected passengers. Survey questions will address average waiting time, preferred bus conditions, willingness to board crowded buses, passenger satisfaction, and reasons for rejecting buses. Responses will be summarised using percentages, charts, and basic statistical comparisons.

3.4 Variables to be Measured

The study will measure the following variables:

- Passenger waiting time;
- Passenger count;
- Crowd density;
- Bus arrival intervals;
- Number of rejected buses;
- Boarding frequency; and
- Peak-hour and off-peak-hour passenger flow.

3.5 Data Recording Procedure

Data collection will be conducted systematically at selected bus stops at different times of day through the following procedure:

1. Visiting selected bus stops during peak and off-peak hours;
2. Observing passenger behaviour directly;
3. Recording passenger counts and waiting times;

4. Conducting short interviews and distributing surveys;
5. Maintaining field notes for each observation session; and
6. Organising collected data into tables and records for analysis.

Multiple observation visits will be conducted at each location to improve accuracy and reliability.

3.6 Group Member Responsibilities

Research tasks will be distributed equitably among the three group members to ensure efficient data collection and coordination.

Member 1 – Observation and Passenger Counting Observe passenger behaviour; count passengers during selected time periods; measure crowd density; and record boarding and rejection behaviour.

Member 2 – Interviews and Survey Collection Conduct informal interviews; distribute and collect survey questionnaires; record passenger opinions and responses; and organise qualitative data.

Member 3 – Waiting Time and Field Documentation Measure passenger waiting times; record bus arrival intervals; maintain field notes; and document environmental conditions and special observations.

All group members will participate in field visits and collaborate during data analysis and report preparation.

Chapter 4 Critical Analysis

This chapter critically discusses the importance of studying passenger waiting behaviour and evaluates the practical challenges and limitations associated with the research. It also highlights the ethical considerations that must be followed during data collection.

4.1 Importance of Studying Passenger Behaviour

Public transportation is central to urban daily life, and passenger behaviour at bus stops directly affects crowd management, boarding efficiency, passenger safety, and the overall quality of the public transport experience. Studying waiting behaviour can reveal how people respond to overcrowding, long waiting times, and irregular bus arrivals, thereby identifying systemic transport problems.

This study is important because it:

- Elucidates real-world commuter decision-making processes;
- Identifies behavioural patterns under crowded conditions;
- Provides insights into passenger comfort and safety concerns;
- Contributes to improving public transportation experiences; and
- Supports the development of more efficient passenger management strategies.

From an academic perspective, the research is valuable because it applies both qualitative and quantitative methods to study real-world human behaviour in public environments, aligning with contemporary mixed-method research traditions (?).

4.2 Expected Challenges During Data Collection

Several challenges are anticipated. Overcrowding during peak hours may make it difficult to observe behaviours accurately, count passengers, and conduct interviews without disruption. The unpredictable nature of bus transportation—including

delays, traffic congestion, and sudden changes in passenger flow—may affect the consistency of observations and waiting-time measurements.

Passengers may also be unwilling to participate in interviews or surveys due to time constraints, privacy concerns, or lack of interest, and some may provide incomplete or inaccurate responses. Environmental conditions such as rain, noise, heat, and traffic congestion may hinder field observations and data recording. Managing multiple concurrent activities—observation, counting, and interviews—will require careful coordination among group members.

4.3 Possible Limitations of the Study

The research is limited to three selected bus stops in Colombo; the observed behavioural patterns may therefore not fully represent passenger behaviour across all bus stops or transport environments in Sri Lanka. The observation period is restricted to selected days and times, meaning that behaviour during weekends, holidays, special events, or atypical transport conditions falls outside the study scope.

Additionally, certain passenger decisions and emotional responses cannot be directly measured through observation alone; some behaviours may be influenced by personal factors that remain invisible during fieldwork. The accuracy of waiting-time measurements and passenger counts may also be affected by heavy crowd movement and rapid changes in passenger flow during peak hours. Despite these limitations, the study is expected to yield meaningful insights into passenger waiting behaviour and boarding decisions in urban transport environments.

4.4 Ethical Considerations

The research will adhere to ethical standards throughout all stages of data collection. Passenger privacy and confidentiality will be respected; no personal information such as names, contact details, or identification will be collected. Informal interviews and surveys will be conducted only with the informed consent of participants, who retain the right to withdraw at any time.

Researchers will avoid disrupting normal passenger activities during observations, which will focus exclusively on general public behaviour in open, publicly accessible spaces. All collected data will be used solely for the academic purposes of this

study, and findings will be reported honestly and accurately without manipulation or misrepresentation.